

SML Assignment 2

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Code Explanation

1. Code snippets for reading data and computing linear and quadratic discriminants using MLE estimates were carried forward from Assignment 1.

PCA

1. Created 784x300 data matrix using numpy reshape and concatenation.
2. Computed mean by summing columns of data matrix and dividing by 300.
3. Centered data matrix by subtracting the mean from each column.
4. Computed unbiased covariance for the centered data matrix $XX^T/(N - 1)$.
5. Solved eigenvalue problem for the given covariance matrix using numpy, and arranged the obtained eigenvectors in non-increasing order of their eigenvalues.
6. Note that the solution of the eigenvalue problem contained eigenvalues which were negative or complex. However, the negative and complex portions of the eigenvalues were negligible, often of the order of 10^{-13} to 10^{-21} , so only positive real parts were taken into consideration.
7. This was mainly due to machine calculation errors, since the covariance matrix is symmetric and positive semidefinite, so the eigenvalues should've been real and positive.
8. Based on the given variance preservation threshold, selected the smallest number of eigenvectors corresponding to the smallest number of (so the largest eigenvalues) which preserved variance.
9. Returned the 2 eigenvectors corresponding to 2 largest eigenvalues for 2 component case.

FDA

1. Computed between class scatter by computing the class means and global mean (average of means of our 3 classes, since each class had an equal size of samples coming from it which is 100 samples), and then computing

$$\sum_{c=1}^3 100 * (u_c - u)(u_c - u)^T.$$

2. Computed within class scatter by using the class-wise biased (MLE) covariances and scaling them appropriately

$$\sum_{c=1}^3 100 * S_c = \sum_{c=1}^3 100 \sum_{i=1}^{100} (x_i - u_c)(x_i - u_c)^T / 100$$

3. To solve the generalised eigenvalue problem, $S_b w = \lambda S_w w$, converted it to regular eigenvalue problem as $S_w^{-1} S_b w = \lambda w$

4. If Sw was not invertible, made it invertible by adding epsilon times the identity matrix (epsilon = 0.001 in the code, any small value works).
5. Obtained 2 eigenvectors for W, since W is 784 x (c - 1), and c = 3 for digits 0,1,2 (3 class case).
6. Again, negative and complex values were adjusted, since we expect the matrix is the regular eigenvalue problem to be semidefinite, and we did not get much difference either, since we picked the 2 largest eigenvalues and corresponding eigenvectors, the rest of the values were close to zero and only 2 values were much larger than the order of 10^{-13} to 10^{-21} .

Finally conducted classification of projected samples, and compared accuracy below. Also included point projection and reconstruction for 6 points (2 of each class) for PCA, and computed mean squared error, which was larger for PCA preserving 75% variance, and smaller (though only by a little bit) for PCA preserving 90% variance, while it was largest for PCA with only 2 components.

Projection and Reconstruction

$$y = U^T(x - u), x' = Uy + u = UU^T(x - u) + u$$

Accuracy Comparison

	FDA		PCA 75% Variance		PCA 90% Variance		PCA 2 Components	
Accuracy	Train Set	Test Set	Train Set	Test Set	Train Set	Test Set	Train Set	Test Set
LDA Accuracy	1.0	0.8933	0.3333	0.3333	0.3333	0.3333	0.3333	0.3333
QDA Accuracy	1.0	0.8933	0.63	0.64	0.66	0.6366	0.38	0.4

Accuracy Order

FDA > PCA 90% > PCA 75% > PCA 2 Components

This is expected because FDA projects the points to retain discriminatory features, while PCA projects to retain variance. PCA results in higher accuracy if higher variance is preserved (90% case), as discriminatory features are not lost, though not by a lot, which is probably because the extra variance is just noise. The first two components of

PCA preserve lesser variance than 75% (took 19 components), so the accuracy is even lower.

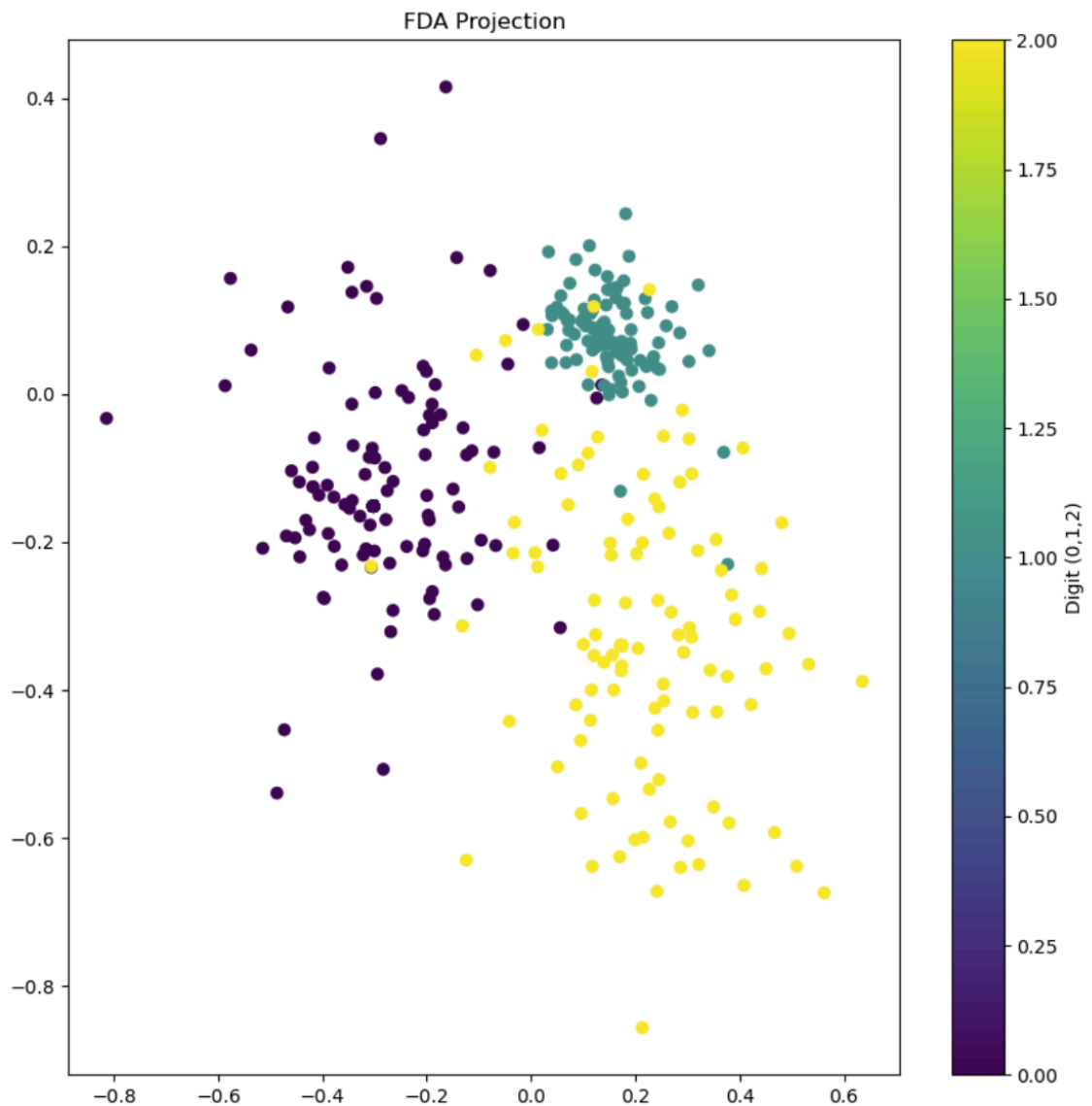
Mean Squared Error of Reconstructed Points

PCA 2 Components > PCA 75% > PCA 90%

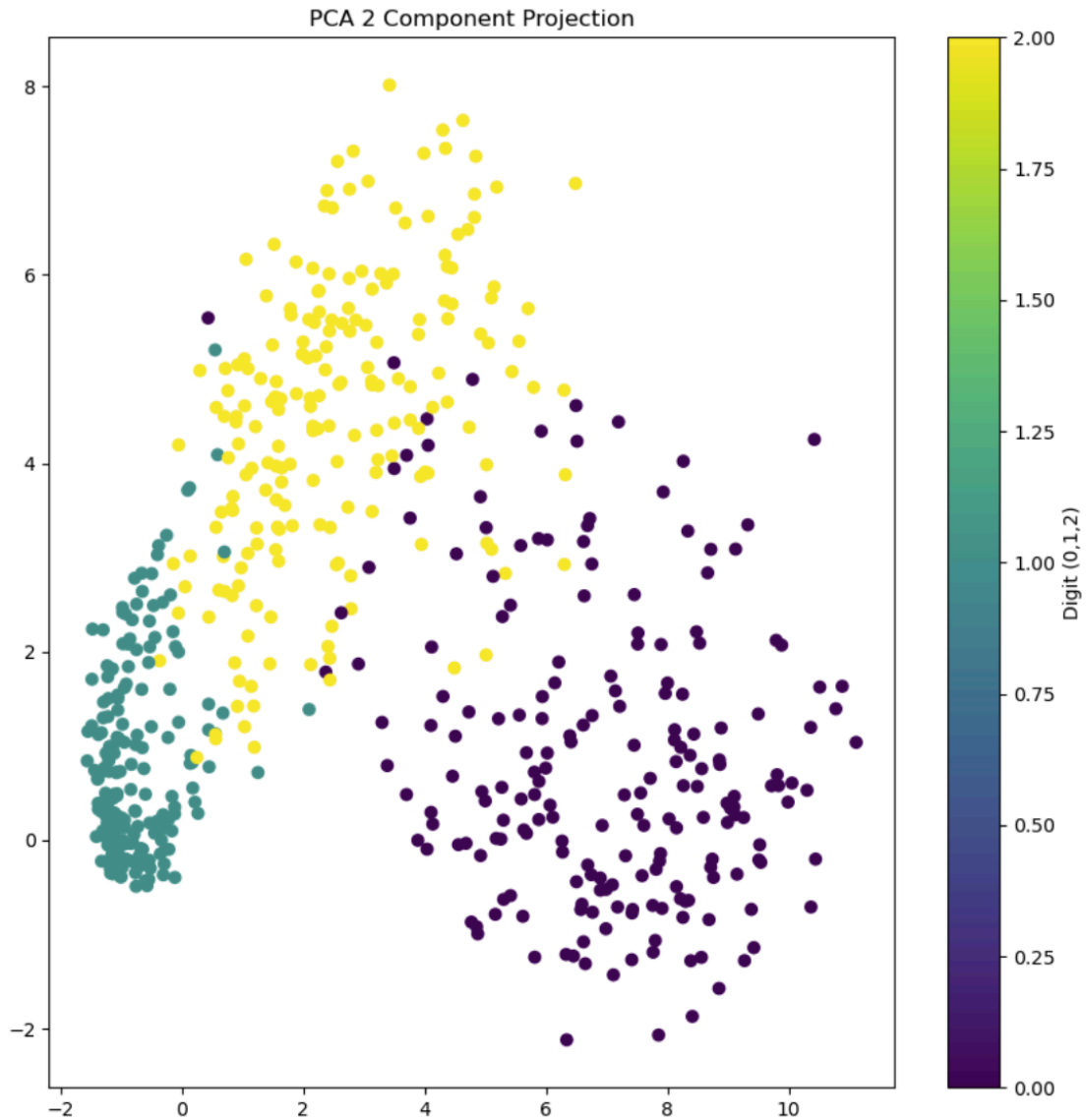
MSE	PCA 75% Variance	PCA 90% Variance	PCA 2 Components
Test	14.54	8.59	34.81
Train	14.56	5.53	33.51

Transformed Feature Space

1) FDA



2) PCA (2 components)



PCA with 2 components projections has a very large spread, while FDA projections are narrower. FDA projections have smaller inter-class overlap regions as compared to PCA projections. They both look like they can be discriminated by appropriate linear and quadratic decision boundaries, though the FDA projections seem more segregated than PCA projections.

Reconstruction After PCA (2 components case)

